

Fady Philips

(709) 219-2119

fadyaphilips97@gmail.com

[Portfolio Website](#)

Windsor, ON | Open to Relocation

EDUCATION

Master of Applied Computing – Internship Stream
University of Windsor — Windsor, ON

Sep 2025 – Dec 2026

Awards: Best Overall Graduate Demo, CS Demo Day Winter 2026 · Ottawa Infotainment Hackathon 2nd Place

Bachelor of Science – Major in Computer Science
Memorial University of Newfoundland — St. John's, NL

Sep 2018 – May 2022

CERTIFICATIONS

Cybersecurity Workshop – Ethical Hacker, Network Defense & Digital Forensics
Sterling Information Technologies / EC-Council

Sep 2025 – Nov 2025

TECHNICAL SKILLS

Languages: JavaScript · TypeScript · Python · C/C++ · Java · Bash · HTML5 · CSS3

Frontend & Mobile: React.js · React Native · Redux · Redux Toolkit · SwiftUI · Expo · Storybook · pygame

Backend & Cloud: Node.js · AWS (Lambda, DynamoDB, API Gateway, Amplify) · Django · Firebase · MySQL · MongoDB · PostgreSQL · MS SQL Server

DevOps & Tooling: Docker · Git · GitHub Actions · CI/CD · Jira · Linux · Unix · Network Security · Selenium

Systems & Embedded: Raspberry Pi · MQTT · OpenCV · Computer Vision · Autonomous Navigation

Design & Architecture: OOP · ECS · MVC · Microservices · REST API · Agile · SCRUM · Figma · UI/UX Design

AI & LLM Tools: Claude Code · OpenAI API · ChatGPT · Prompt Engineering · RAG · AI Agents · MCP

EXPERIENCE

Private Python Instructor, Remote

Nov 2024 – Present

Tech: Python, Scratch, Curriculum Design

- Teach Python and Scratch constructs: control flow, iteration, functions, and data structures
- Devise hands-on coding exercises to strengthen logical reasoning and problem decomposition
- Emphasize debugging, code clarity, and structured problem solving

Frontend Developer — Yekola, Toronto

Oct 2023 – Oct 2024

Tech: React Native, Redux, UML, JavaScript, Agile, Team Leadership

- Led 4 frontend devs at an early-stage EdTech startup building a language learning platform
- Spearheaded early technical decisions on stack and architecture
- Collaborated in Agile sprints to define features, review code, and align UI with product goals
- Designed app flow and UX with UML, then built the mobile frontend using React-Native and Redux

Frontend Developer — NLEats, St. John's

May 2020 – Dec 2020

Tech: React.js, JavaScript, Responsive Design, Agile, Mentorship

- Mentored 3 new hires and volunteers on project's tech stack and provided resources
- Developed and tested responsive web interfaces using React.js, ensuring clean component design
- Participated in sprint planning, code reviews, and backlog refinement within an Agile team

SELECTED PROJECTS

[PortfolioOS: Interactive Windows 98 Portfolio — Personal Project](#)

Tech: React.js, Vite, JavaScript, Vitest, CSS, Claude Code

- Came up with the idea for a portfolio presented as a working Windows 98 desktop, and designed it
- Built a window manager with dragging, 8-direction resizing, minimizing and a taskbar
- Designed a single list of programs that drives the icons, the Start menu and the windows
- Kept all portfolio text in JSON, so updating the content needs no code changes
- Built themed programs, including a file explorer, a contact form that really sends email, and Solitaire
- Used Claude Code skills to go from a written design spec to a plan to test-driven code
- Reviewed every AI-generated change for architecture, performance and accessibility before merging

LancerFit: UWindsor Gamified Fitness Platform — Team Project

Tech: React Native, Expo, Node.js, TypeScript, PostgreSQL, Docker, Claude Code

- Led UI/UX design and prototyping in Figma, delivering wireframes through stakeholder approval to a working HTML prototype
- Designed and built the badge engine end-to-end across backend and frontend, supporting five rule families with automated evaluation
- Contributed to mobile UI implementation across multiple screens using React Native and Expo
- Leveraged Claude Code and AI-assisted workflows throughout development, treating generated output as proposals requiring architectural review

Autonomous 4WD Raspberry Pi Smart Car — Team Project

Tech: Python, Raspberry Pi, MQTT, OpenCV, pygame, Computer Vision, Autonomous Navigation

- Architected a real-time MQTT communications backend relaying motor commands, telemetry, and mode switches between a Raspberry Pi and laptop over local WiFi at sub-10ms latency
- Developed an OpenCV overhead camera pipeline for live position and heading tracking
- Implemented autonomous waypoint navigation with obstacle avoidance and stuck recovery
- Streamed live onboard video to a pygame controller UI with servo and autonomous mode controls
- Won Best Overall Graduate Demo at University of Windsor Winter 2026 CS Demo Day

Neutron React Game Engine — Solo Project (In Progress)

Tech: React.js, JavaScript, Redux Toolkit, ECS, OOP, Performance Optimization

- Built a Bomberman clone in React.js with game logic designed around reusable ECS components
- Managed global game state using Redux Toolkit with a focus on performance optimization
- Architected across three phases: game implementation, logic extraction, and engine-driven rebuilding

AI HTML Generator — Solo Project

Tech: React.js, JavaScript, OpenAI API, Firebase, CI/CD

- Built a React.js app using the OpenAI API to generate HTML from natural-language prompts
- Implemented an automated CI/CD pipeline to deploy application to Firebase
- Completed as part of a 3-hour mini hackathon

My-Flix: V2 — Team Project

Tech: React, TypeScript, AWS Amplify, AWS Lambda, AWS Serverless, Ant Design, Figma, RESTful APIs

- Web app built with React, AWS Amplify, and AWS SAM to help users discover their next movie
- Migrated MERN backend to AWS Serverless, cutting infrastructure costs 90% via event-driven compute
- Built high-quality UI using React, TypeScript, Ant Design, and Figma
- Integrated RESTful APIs with AWS Serverless, working with JSON data models

PR Automation Pipeline — Solo Project

Tech: Docker, GitHub Actions, CI/CD, Bash

- Created Pipeline to generate a docker file for every PR allowing QA to test different branches simultaneously in containerized environments
- Utilizes GitHub Artifacts for storing branch associated docker files

GymTheory — Solo Project

Tech: React Native, Expo, AWS DynamoDB, AWS Lambda, Microservices

- Built a mobile app to log daily workouts using React Native and Expo for cross-platform development
- Implemented a backend in AWS DynamoDB and AWS Lambda with a Microservices architecture

Systems Programming Projects — University of Windsor

Tech: C, Linux, Unix, Bash, Socket Programming, POSIX

- Built a distributed client-server file system in C with multi-client support via socket programming
- Implemented a Mini Bash shell from scratch in C using fork(), exec(), pipes, FIFO, and I/O redirection
- Developed a process tree analyzer in C supporting 15+ signal and memory inspection operations

Space Wars — Team Project

Tech: C++, SFML, OOP, ECS

- Developed a game in Intro to Game Development, coordinating tasks across a 4-person team
- Game used C++ and the SFML library with Object Oriented Design and ECS for architecture

RECENT HACKATHONS

Ottawa Infotainment UI/UX Hackathon— Windsor, ON

Jan 2026 - Mar 2026

Tech: SwiftUI, Figma, iOS Development, UI/UX Design, UnityUI

- Designed and developed an iPad car dashboard in SwiftUI as part of a 4-person team, placing 2nd
- Prototyped and iterated on the dashboard UI in Figma before implementation
- Selected as 1 of 15 participants across all teams for a week-long internship opportunity